

Identifying Post-COVID Risk Factors with Model-Agnostic Feature Importance

using the `xplainfi` R package on the German national cohort (NAKO)

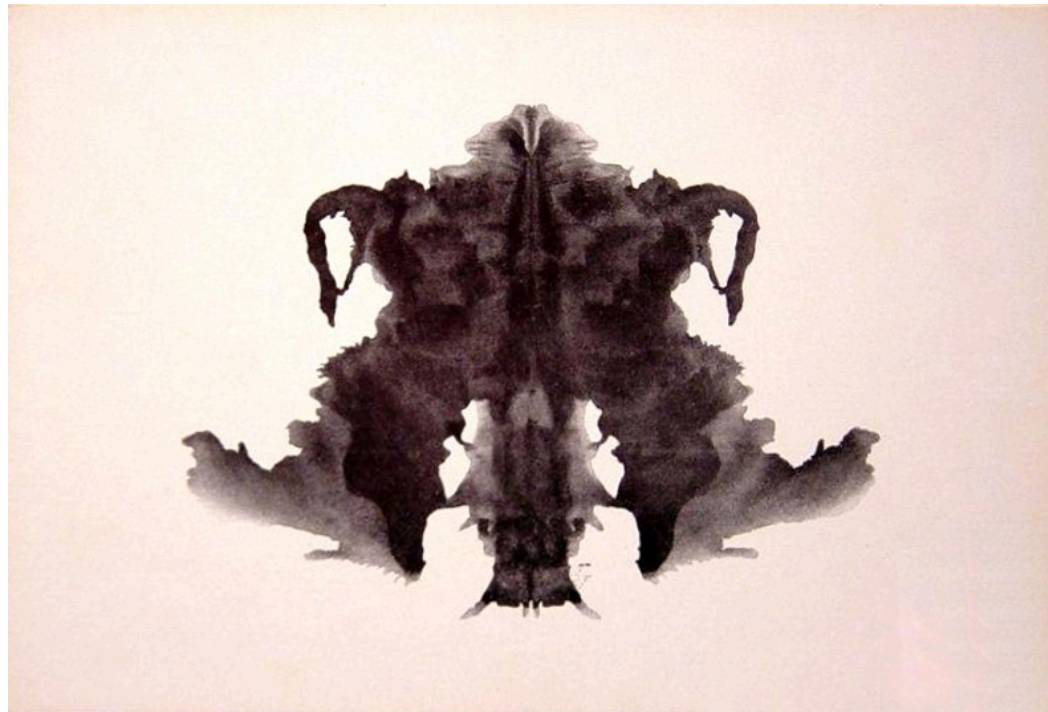
Lukas Burk

Leibniz Institute for Prevention Research and Epidemiology — BIPS
Bremen, Germany

2026-05-19

CEN 2026

A matter of interpretation



Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)
- Working definition: (Mikolajczyk et al., 2024)

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)
- Working definition: (Mikolajczyk et al., 2024)
 - “Any” PCC: at least **1** symptom persistent **4+ months after** infection

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)
- Working definition: (Mikolajczyk et al., 2024)
 - “Any” PCC: at least **1** symptom persistent **4+ months after** infection
 - “severe” PCC: at least **9**

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)
- Working definition: (Mikolajczyk et al., 2024)
 - “Any” PCC: at least **1** symptom persistent **4+ months after** infection
 - “severe” PCC: at least **9**
- Many open questions: risk factors, causal factors, subgroups?

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
 - Clinically heterogenous (symptoms & severity)
 - Working definition: (Mikolajczyk et al., 2024)
 - “Any” PCC: at least **1** symptom persistent **4+ months after** infection
 - “severe” PCC: at least **9**
 - Many open questions: risk factors, causal factors, subgroups?
- **RESOLVE-PCC** project funded by the German BMFTR

Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)
- Working definition: (Mikolajczyk et al., 2024)
 - “Any” PCC: at least **1** symptom persistent **4+ months after** infection
 - “severe” PCC: at least **9**
- Many open questions: risk factors, causal factors, subgroups?

→ **RESOLVE-PCC** project funded by the German BMFTR

Federal Ministry of
Research, Technology and Space

Why feature importance?

What it can and can't tell us about PCC



3

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?

Why feature importance?

What it can and can't tell us about PCC



3

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?
- Applications:
 - Feature selection 🤝 feature importance

Why feature importance?

What it can and can't tell us about PCC

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?
- Applications:
 - Feature selection 🤝 feature importance
 - Sanity check: is the model relying on plausible signal?

Why feature importance?

What it can and can't tell us about PCC

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?
- Applications:
 - Feature selection 🤝 feature importance
 - Sanity check: is the model relying on plausible signal?
 - Hypothesis generation

Why feature importance?

What it can and can't tell us about PCC

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?
- Applications:
 - Feature selection 🤝 feature importance
 - Sanity check: is the model relying on plausible signal?
 - Hypothesis generation
 - **FI** $\neq$ **causal effect** but associations still informative (Ewald et al., 2024)
- “Importance” itself is not a single quantity

Why feature importance?

What it can and can't tell us about PCC

- Predictive models use many candidate risk factors
- Which ones actually drive predictions?
- Applications:
 - Feature selection 🤝 feature importance
 - Sanity check: is the model relying on plausible signal?
 - Hypothesis generation
 - **FI** $\neq$ **causal effect** but associations still informative (Ewald et al., 2024)
- “Importance” itself is not a single quantity $\rightarrow$ *method choice* defines the answer

Common FI methods

How “important” is feature X_j ?



Common FI methods

How “important” is feature X_j ?



4

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Leave-one-covariate-out (**LOCO**)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Perturbation of X_j

1. Fit full model
2. Measure performance on...

Leave-one-covariate-out (**LOCO**)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Perturbation of X_j

1. Fit full model
2. Measure performance on...
 - a) all test data

Leave-one-covariate-out (**LOCO**)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Perturbation of X_j

1. Fit full model
2. Measure performance on...
 - a) all test data
 - b) same data where X_j is *randomly permuted*
3. Compare performance

Leave-one-covariate-out (*LOCO*)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Perturbation of X_j

1. Fit full model
2. Measure performance on...
 - a) all test data
 - b) same data where X_j is *randomly permuted*
3. Compare performance
4. (Repeat k times for stability)

Leave-one-covariate-out (*LOCO*)

Common FI methods

How “important” is feature X_j ?

Refitting without feature X_j

1. Fit full model
2. Fit model without X_j
3. Compare performance
4. (Repeat k times for stability)

Leave-one-covariate-out (**LOCO**)

Perturbation of X_j

1. Fit full model
2. Measure performance on...
 - a) all test data
 - b) same data where X_j is *randomly permuted*
3. Compare performance
4. (Repeat k times for stability)

Permutation feature importance (**PFI**)

Conditional FI

Permute, but respect feature dependence



5

- PFI permutes X_j marginally $\rightarrow$ implausible combinations (Hooker et al., 2021)
- E.g.: “20-year-old with 30 years smoking history”

Conditional FI

Permute, but respect feature dependence



5

- PFI permutes X_j marginally $\rightarrow$ implausible combinations (Hooker et al., 2021)
- E.g.: “20-year-old with 30 years smoking history”
- CFI: perturbation **conditional** on other features

Conditional FI

Permute, but respect feature dependence

5

- PFI permutes X_j marginally $\rightarrow$ implausible combinations (Hooker et al., 2021)
- E.g.: “20-year-old with 30 years smoking history”
- CFI: perturbation **conditional** on other features
- Requires *conditional sampling* $\tilde{X}_j \sim F_{X_j | X_{-j}}$, some options:

Conditional FI

Permute, but respect feature dependence

5

- PFI permutes X_j marginally $\rightarrow$ implausible combinations (Hooker et al., 2021)
- E.g.: “20-year-old with 30 years smoking history”
- CFI: perturbation **conditional** on other features
- Requires *conditional sampling* $\tilde{X}_j \sim F_{X_j | X_{-j}}$, some options:
 - Conditional Gaussian $\rightarrow$ fast, but only continuous data

Conditional FI

Permute, but respect feature dependence

- PFI permutes X_j marginally $\rightarrow$ implausible combinations (Hooker et al., 2021)
- E.g.: “20-year-old with 30 years smoking history”
- CFI: perturbation **conditional** on other features
- Requires *conditional sampling* $\tilde{X}_j \sim F_{X_j | X_{-j}}$, some options:
 - Conditional Gaussian $\rightarrow$ fast, but only continuous data
 - Adversarial random forests (ARF)
 - $\rightarrow$ handles *mixed* data, *missing values*, computationally more expensive

What are we explaining?

Fixed model, learning algorithm, data-generating process?

- **Model-level:** explain a single fitted model
 - One or many permutations (**PFI/CFI**) on a held-out set

What are we explaining?

Fixed model, learning algorithm, data-generating process?

- **Model-level:** explain a single fitted model
 - One or many permutations (**PFI/CFI**) on a held-out set
- **Learner-level FI:** explain the *prediction method*, not one fit
 - **LOCO** automatically refits, includes learner variability

What are we explaining?

Fixed model, learning algorithm, data-generating process?

- **Model-level:** explain a single fitted model
 - One or many permutations (**PFI/CFI**) on a held-out set
- **Learner-level FI:** explain the *prediction method*, not one fit
 - **LOCO** automatically refits, includes learner variability
 - **Any method:** Repeat across resampling (e.g. 15 subsampling iters)
 - Captures variability from data sampling and learner stochasticity

Analysis dataset: NAKO

German National Cohort



7

- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)

Analysis dataset: NAKO

German National Cohort



7

- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)
- $\rightarrow$ Goal: Predict “any PCC” vs. “no PCC”

Analysis dataset: NAKO

German National Cohort



7

- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)
- $\rightarrow$ Goal: Predict “any PCC” vs. “no PCC”
- Mix of *survey* + *in-person assessment*:
 - demographics, anthropometrics, socioeconomic
 - comorbidities, smoking history, biomarker labs (where available)
 - mental-health questionnaires

Analysis dataset: NAKO

German National Cohort



7

- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)
- $\rightarrow$ Goal: Predict “any PCC” vs. “no PCC”
- Mix of *survey* + *in-person assessment*:
 - demographics, anthropometrics, socioeconomic
 - comorbidities, smoking history, biomarker labs (where available)
 - mental-health questionnaires
- ≈ 150 features used (dropping near-constant / near-fully-missing)
- Rare-event indicators retained despite low prevalence

Analysis: 'Any PCC' prediction

Preliminary, exploratory



8

- Learners: Gradient boosting (XGBoost), random forest (ranger)
- Tuned on **PR-AUC** (baseline = 24%)
- General performance:
 - PR-AUC $\approx$ 43%
 - ROC-AUC $\approx$ 64%

Analysis: 'Any PCC' prediction

Preliminary, exploratory



8

- Learners: Gradient boosting (XGBoost), random forest (ranger)
- Tuned on **PR-AUC** (baseline = 24%)
- General performance:
 - PR-AUC $\approx$ 43%
 - ROC-AUC $\approx$ 64%
- **PFI**, **CFI** (+ARF), **LOCO** computed via `xplainfi`

Analysis: 'Any PCC' prediction

Preliminary, exploratory

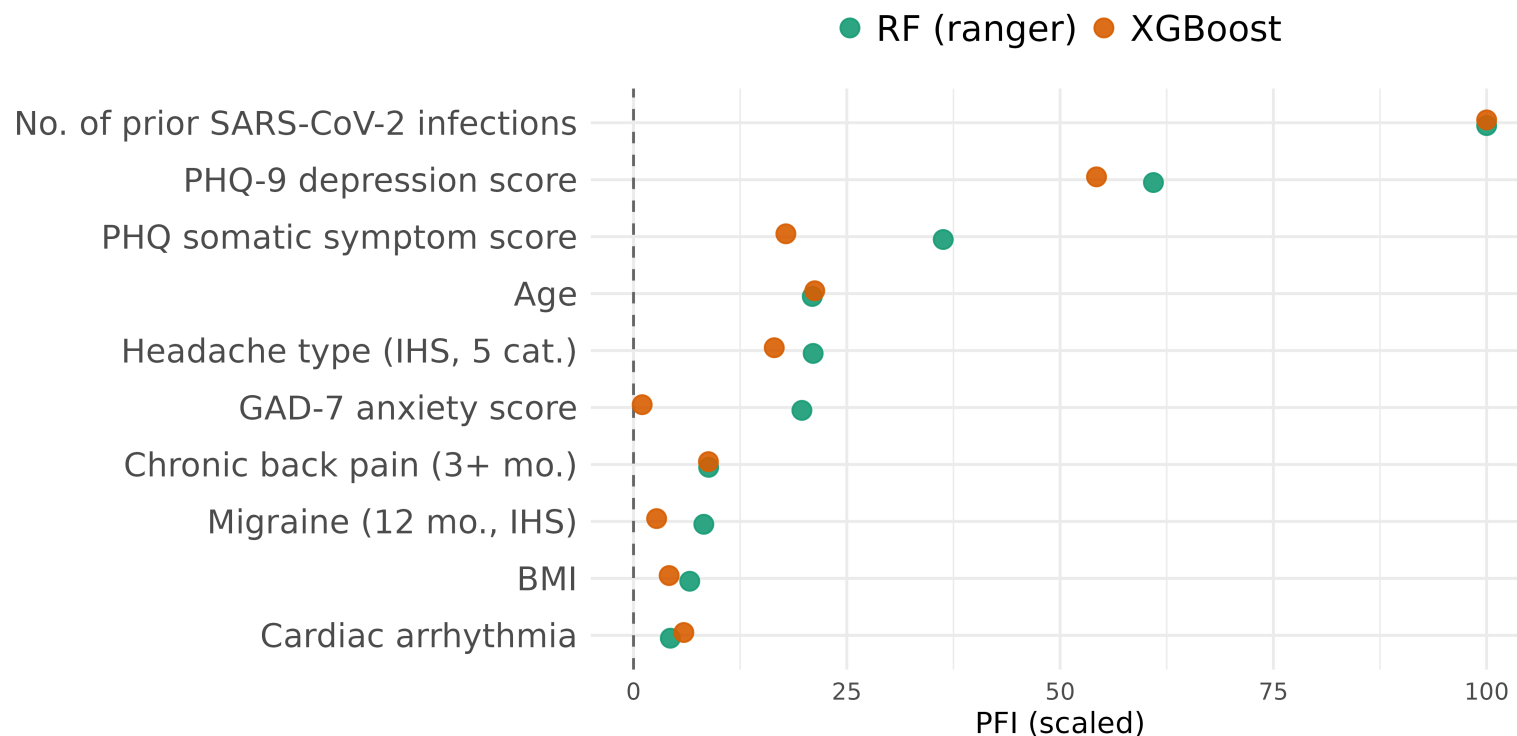


8

- Learners: Gradient boosting (XGBoost), random forest (ranger)
- Tuned on **PR-AUC** (baseline = 24%)
- General performance:
 - PR-AUC $\approx$ 43%
 - ROC-AUC $\approx$ 64%
- **PFI**, **CFI** (+ARF), **LOCO** computed via `xplainfi`
- FI here on test set (i.e. **model importance**)

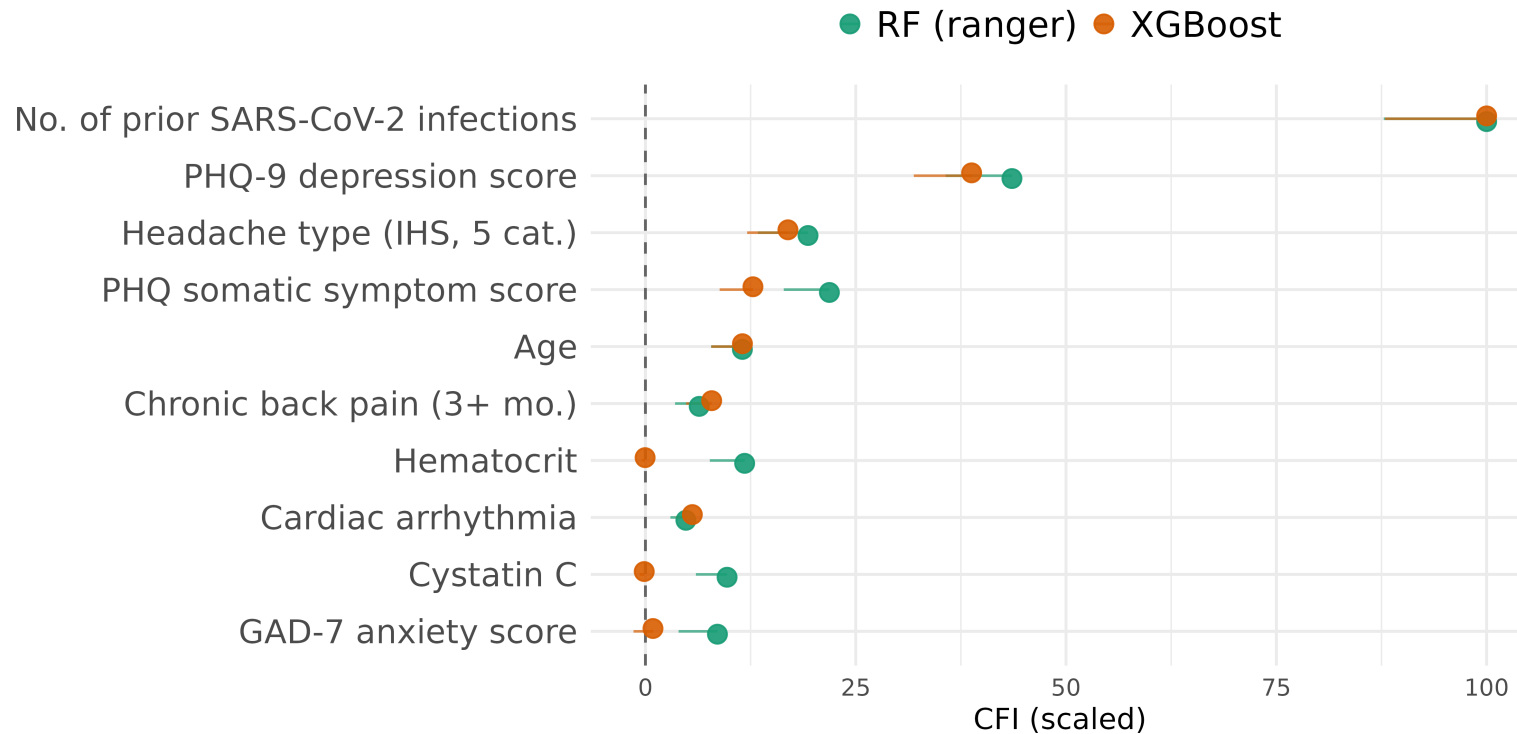
Results: Permutation Feature Importance (PFI)

Top 10 features (between RF + XGBoost)



Results: Conditional Feature Importance (CFI)

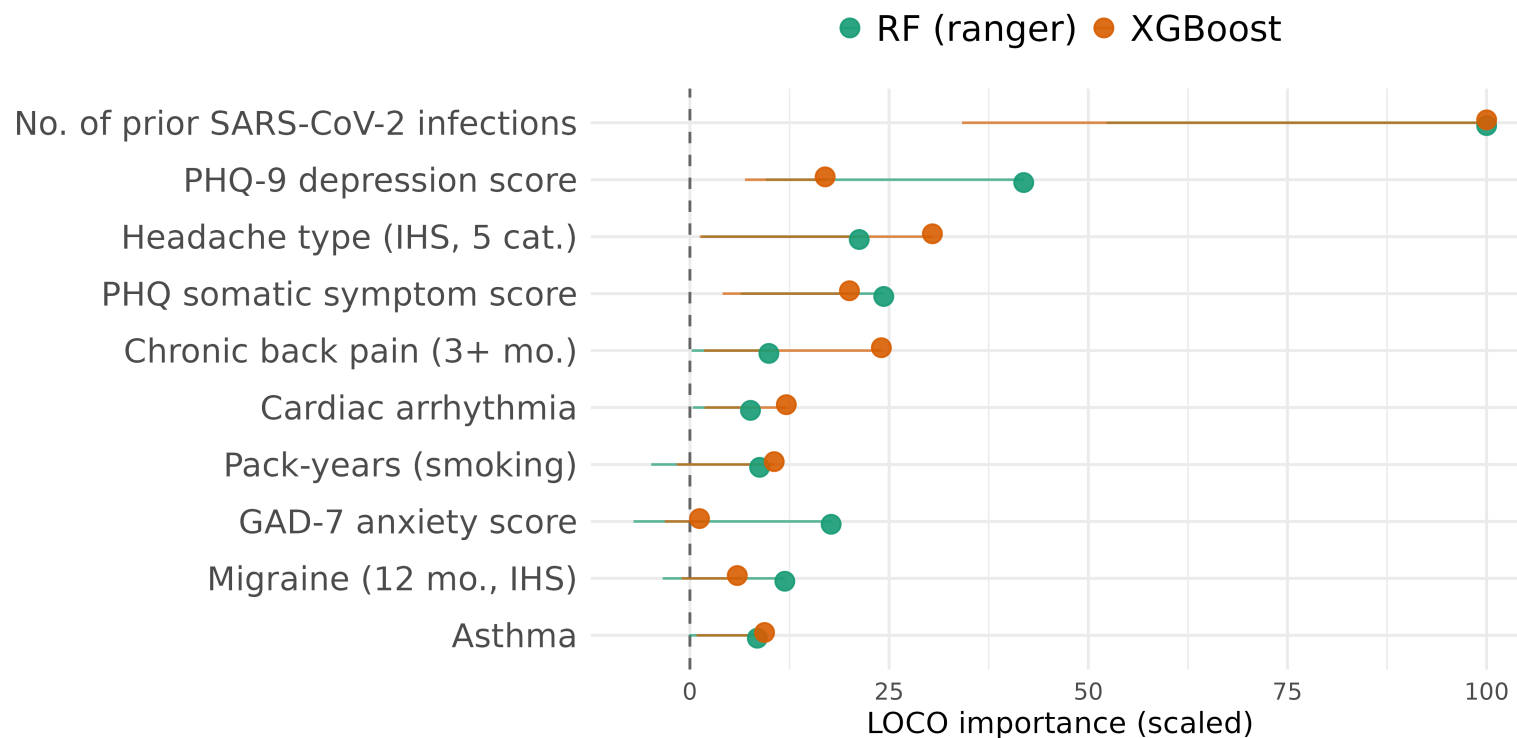
Top 10 features, one-sided 95% CIs



Results: Leave-one-covariate-out (LOCO)

Top 10 features, one-sided 95% CIs

11



Conclusion

Everything is complicated, always



12

- Feature importance is useful, but **not a single number**
- Each method answers a slightly different question
 - **PFI**: marginal, model-faithful but extrapolation issue
 - **CFI**: conditional, more faithful to joint distr., sampling-dependent
 - **LOCO**: refit-based, expensive but assumption-light

Conclusion

Everything is complicated, always

- Feature importance is useful, but **not a single number**
- Each method answers a slightly different question
 - **PFI**: marginal, model-faithful but extrapolation issue
 - **CFI**: conditional, more faithful to joint distr., sampling-dependent
 - **LOCO**: refit-based, expensive but assumption-light
- Compare methods → robustness check, not contradiction
- `xplainfi` provides a unified interface for all of the above (Burk et al., 2026)

Thank you for your attention!



Contact

Lukas Burk

Leibniz Institute for Prevention Research
and Epidemiology – BIPS
Achterstraße 30
28359 Bremen
Germany

burk@leibniz-bips.de



References



13

- Appel, K. S., Nürnberger, C., Bahmer, T., Förster, C., Polidori, M. C., Kohls, M., Kraus, T., Hettich-Damm, N., Petersen, J., Blaschke, S., Bröhl, I., Butzmann, J., Dashti, H., Deckert, J., Dreher, M., Fiedler, K., Finke, C., Geisler, R., Hanses, F., ... NAPKON Study Group. (2024). Definition of the Post-COVID Syndrome Using a Symptom-Based Post-COVID Score in a Prospective, Multi-Center, Cross-Sectoral Cohort of the German National Pandemic Cohort Network (NAPKON). *Infection*, 52(5), 1813–1829. <https://doi.org/10.1007/s15010-024-02226-9>
- Burk, L., Ewald, F. K., Casalicchio, G., Wright, M. N., & Bischl, B. (2026, March 16). *Xplainfi: Feature Importance and Statistical Inference for Machine Learning in R*. <https://doi.org/10.48550/arXiv.2603.15306>
- Ewald, F. K., Bothmann, L., Wright, M. N., Bischl, B., Casalicchio, G., & König, G. (2024). A Guide to Feature Importance Methods for Scientific Inference. In L. Longo, S. Lapuschkin, & C. Seifert (Eds.), *Explainable Artificial Intelligence: Explainable Artificial Intelligence*. https://doi.org/10.1007/978-3-031-63797-1_22
- Hooker, G., Mentch, L., & Zhou, S. (2021). Unrestricted Permutation Forces Extrapolation: Variable Importance Requires at Least One More Model, or There Is No Free Variable Importance. *Statistics and Computing*, 31(6), 82. <https://doi.org/10.1007/s11222-021-10057-z>
- Mikolajczyk, R., Diexer, S., Klee, B., Pfrommer, L., Purschke, O., Fricke, J., Ahnert, P., Gabrysch, S., Gottschick, C., Bohn, B., Brenner, H., Buck, C., Castell, S., Gastell, S., Greiser, K. H., Harth, V., Heise, J.-K., Holleczeck, B., Kaaks, R., ... Karch, A. (2024). Likelihood of Post-COVID Condition in People with Hybrid Immunity; Data from the German National Cohort (NAKO). *The Journal of Infection*, 89(2), 106206. <https://doi.org/10.1016/j.jinf.2024.106206>